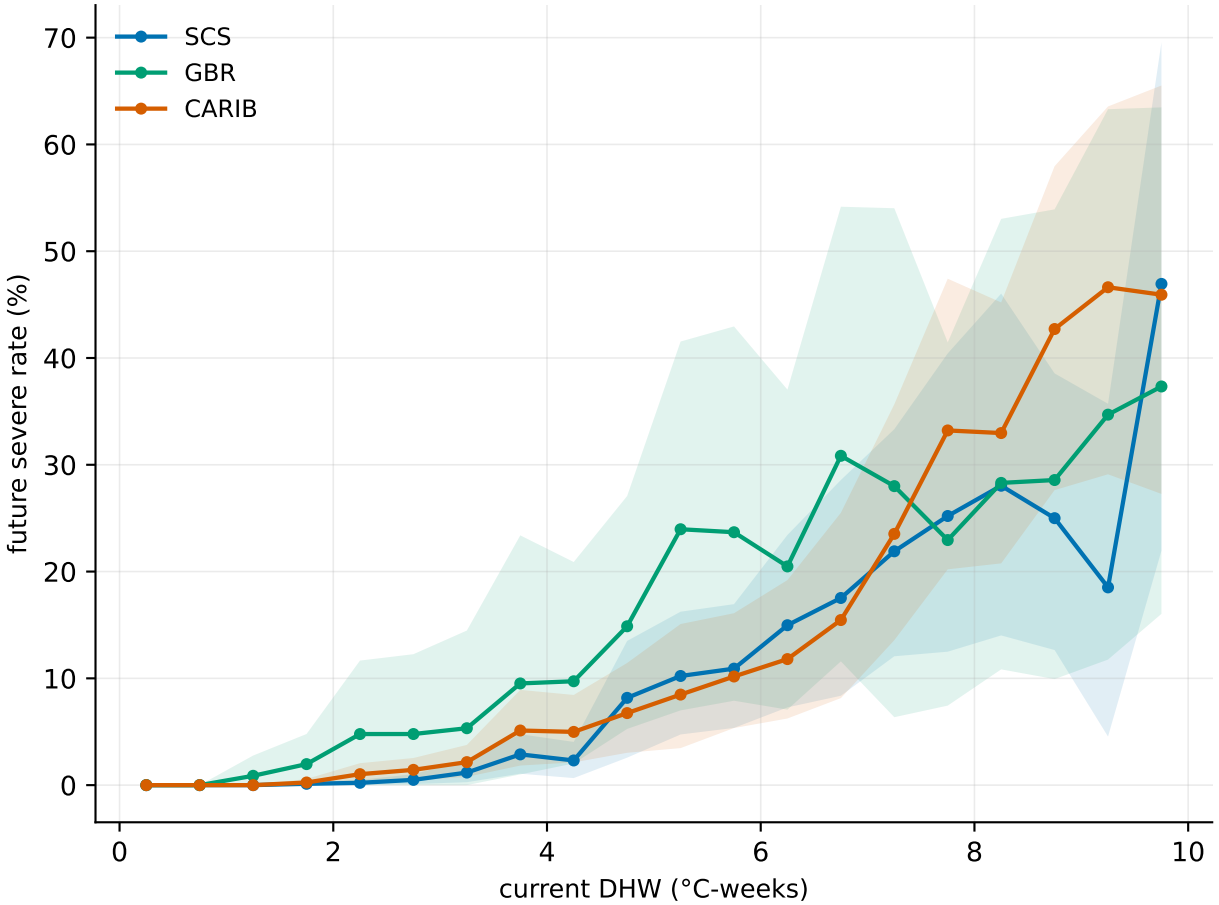
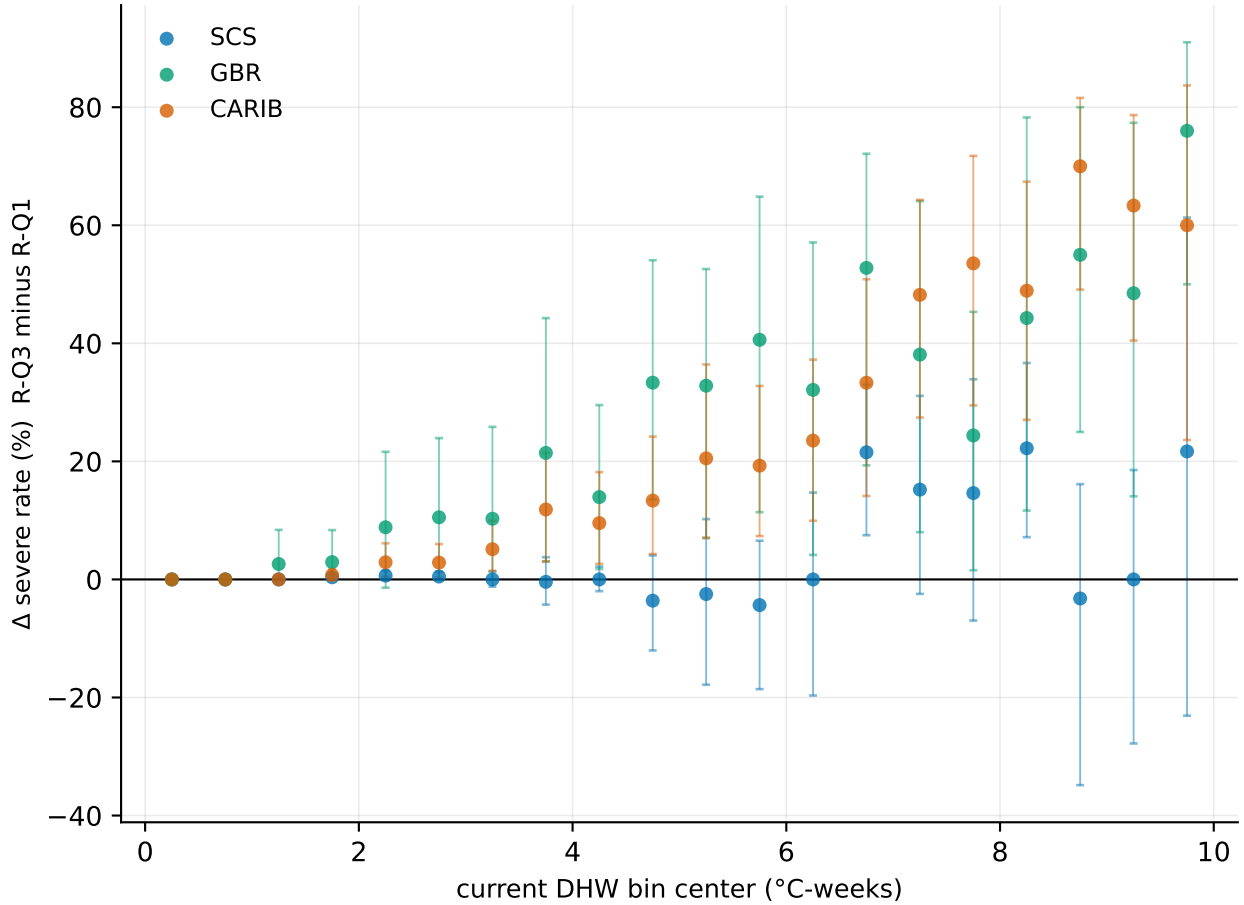
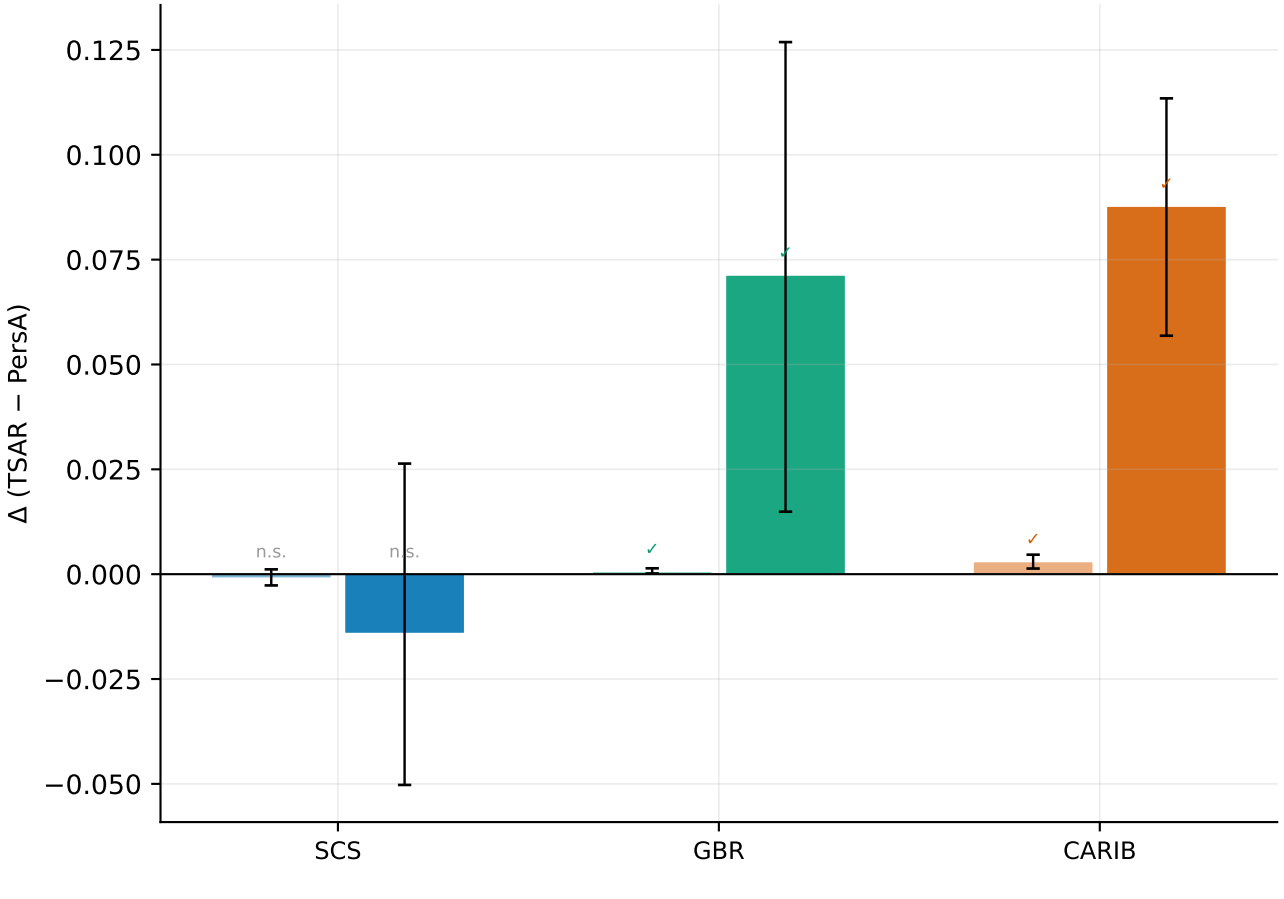
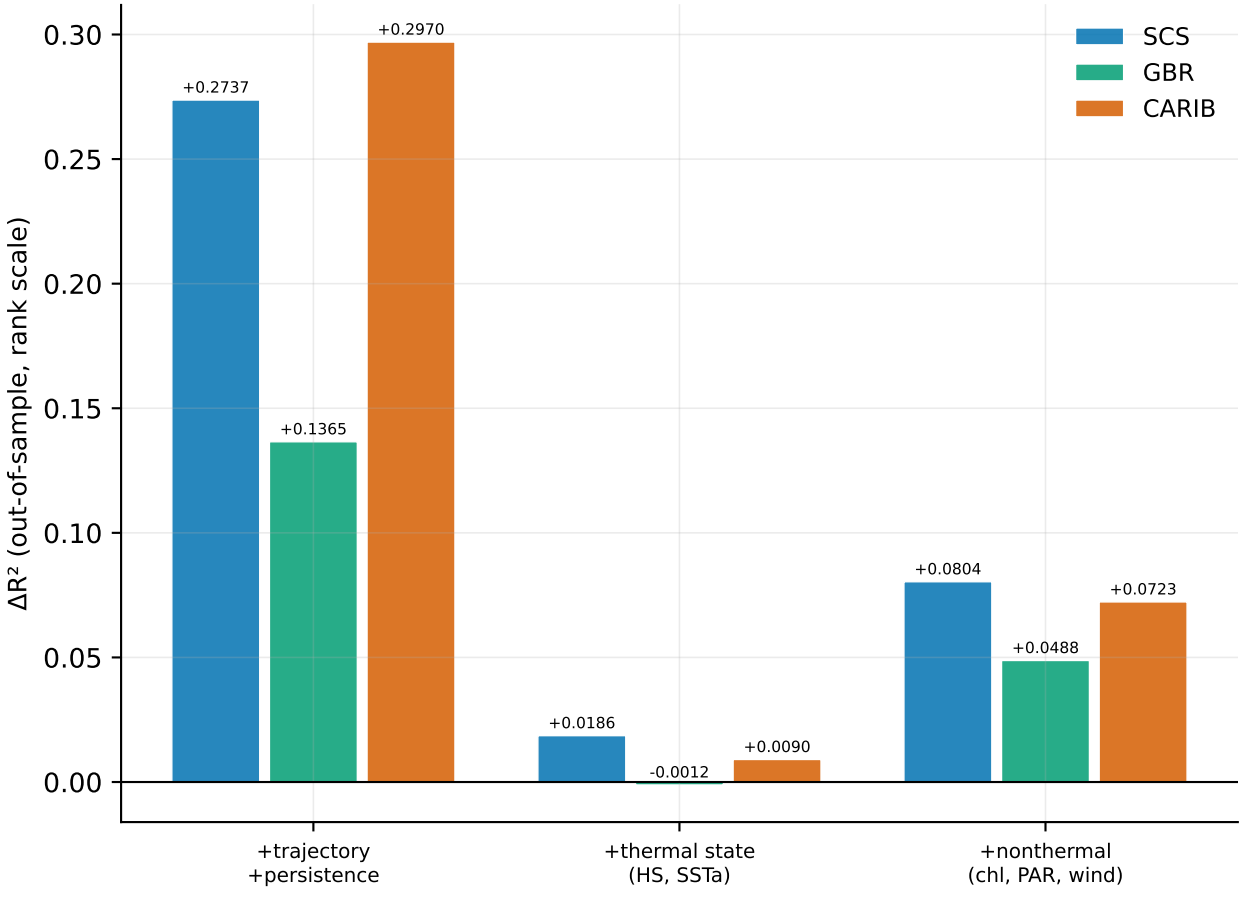
(a) Conditional thermal-response surface
(week-block bootstrap 95% CI)



(b) Residual R separates conditional risk at matched DHW
(CI-excluded points = basin-specific)



(c) Basin-specific conditional-risk increment (ΔROC / ΔPR)
✓ = CI excludes zero; faded = ΔPR



(d) Residual mechanism hierarchy (OOS ΔR²)
M0: DHW → +trajectory → +state → +context